CHOOSING A P-POWER EXPONENT THAT BRINGS A SUM OF TWO VARIABLES INTO A PRESCRIBED SUBRING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Binomial coefficients with p-power valuation. Estimated in the counted proof at lines 11052-11060. Provides the divisibility that membership in the prescribed subring requires.
- Division of exponents by p^n with remainder. Used at proof lines 11044-11046. Splits each monomial exponent into a multiple of p^n or p^m plus a remainder, which is what the divisibility condition is stated in terms of.

Lemma 0.1. *Let p be a prime number. Let $n, m > 0$ be two integers. There exists an integer a such that $(x+y)^{p^a}, p^a(x+y) \in \mathbf{Z}[x^{p^n}, p^n x, y^{p^m}, p^m y]$.*

Proof. This is clear for $p^a(x+y)$ as soon as $a \geq n, m$. In fact, pick $a \gg n, m$. Write

$$(x+y)^{p^a} = \sum\nolimits_{i,j \geq 0, i+j=p^a} \binom{p^a}{i,j} x^i y^j$$

For every $i, j \geq 0$ with $i + j = p^a$ write $i = qp^n + r$ with $r \in \{0, \ldots, p^n - 1\}$ and $j = q'p^m + r'$ with $r' \in \{0, \ldots, p^m - 1\}$. The condition $(x+y)^{p^a} \in \mathbf{Z}[x^{p^n}, p^n x, y^{p^m}, p^m y]$ holds if

$$p^{nr+mr'} \text{ divides } \binom{p^a}{i,j}$$

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If $r = r' = 0$ then the divisibility holds. If $r \neq 0$, then we write

$$\binom{p^a}{i, j} = \frac{p^a}{i}\binom{p^a - 1}{i - 1, j}$$

Since $r \neq 0$ the rational number p^a/i has p-adic valuation at least $a-(n-1)$ (because i is not divisible by p^n). Thus $\binom{p^a}{i,j}$ is divisible by p^{a-n+1} in this case. Similarly, we see that if $r' \neq 0$, then $\binom{p^a}{i,j}$ is divisible by p^{a-m+1}. Picking $a = np^n + mp^m + n + m$ will work. $\square$